

Nisha Chaudhary

PhD, Jamia Millia Islamia

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AI scientist with a PhD in computational biology and expertise in digital pathology, deep learning, and cancer diagnostics. Experienced in building clinically grounded AI systems for tumor detection, grading, and prognosis, with a strong focus on large-scale WSI datasets, model interpretability, and translational relevance. Actively working toward multimodal integration of histopathology with spatial and molecular data for precision oncology.

Education

Jamia Millia Islamia

PhD Computational Biology

Advisor: Dr. Tanveer Ahmad

New Delhi, IN

Feb. 2025

Jamia Millia Islamia

M.Sc. Bioinformatics

New Delhi, IN

2015 - 2017

Maharshi Dayanand University

B.Sc. (Hons) Zoology

Haryana, IN

2012 - 2015

Professional Experience

Cellogen Therapeutics Pvt. Ltd

Senior computational scientist (Research & Development, AI ML Unit)

Noida, U.P., IN

2025-Present

Applying LLM-based and deep learning approaches to biological sequence modeling for Cas enzyme discovery. Curating and analyzing large-scale CRISPR-associated protein sequence datasets. Developing reproducible computational workflows for sequence representation learning and functional prediction. Supporting translational genome-engineering research through computational modeling.

MicroCRISPR Pvt. Ltd

Manager (Research & Development, AI ML Unit)

New Delhi, IN

2024-2025

- ❖ Led development of AI/ML pipelines for CRISPR-based diagnostic and therapeutic applications.
- ❖ Coordinated model development, data curation, and evaluation workflows in an industry R&D setting.
- ❖ Bridged computational modeling with experimental and translational objectives.

Jamia Millia Islamia

Advisor: Dr. Tanveer Ahmad

New Delhi, IN

2018-2024

• Ph.D. Scholar

Artificial intelligence based oral cancer diagnosis and prognosis using digital pathology images

- ❖ Designed and developed a dual-stage weakly supervised MIL framework for tumor detection and

grade-level subtyping of oral squamous cell carcinoma (OSCC) from whole-slide images.

- ❖ Worked with large multi-institutional WSI cohorts, including data curation, patch extraction, annotation coordination, and quality control.
- ❖ Implemented explainable AI methods using attention heatmaps and patch-level interpretability to support pathologist trust and clinical relevance.
- ❖ Led end-to-end model development: preprocessing, feature extraction, training, validation, and statistical evaluation.
- ❖ Collaborated closely with oral pathologists and clinicians to align AI outputs with diagnostic workflows and grading criteria.

Jamia Hamdard

New Delhi, IN

Advisor: Dr. Mymoona Akhtar

2017 (six months)

- Studentship, DBT-funded Bioinformatics Centre (BIC)
Pharmacophore mapping of newly patented antimalarial agents and virtual screening of database(s) opt in newer antimalarial agents

Jamia Millia Islamia

New Delhi, IN

Advisor: Dr. Mohammad Husain

2016 (one month)

- Summer intern, Centre for Innovation and Entrepreneurship
In-silico Designing of Anticancer Drug for MMPs family

Publications

* first author

- **Chaudhary N***, Muddemanavar P, Singh DK, Rai A, Mishra D, Sowmya SV, Augustine J, Chandra A, Chaurasia A, Ahmad T. *Dual-stage AI system for Pathologist-Free Tumor Detection and subtyping in Oral Squamous Cell Carcinoma*. [medRxiv](#). 2025 Jun 5; 2025.06.05.25329090.
- **Chaudhary N***, Rai A, Rao AM, Faizan MI, Augustine J, Chaurasia A, Mishra D, Chandra A, Chauhan V, Ahmad T. *High-resolution AI image dataset for diagnosing oral submucous fibrosis and squamous cell carcinoma*. [Scientific Data](#). 2024 Sep 27;11(1):1050.
- Ansari MS*, Chauhan V, Singh A, Akhtar A, **Chaudhary N**, Tyagi R, Divya, Husain K, Sharma S, Alam R, Shakir M. *AI-guided CAR designs and targeted pathway modulation to enhance multi-antigen CAR T cell durability and overcome antigen escape*. [Nature Communications](#). 2026 Jan 17.
- Sagar S*, Faizan MI, **Chaudhary N**, Singh V, Singh P, Gheware A, Sharma K, Azmi I, Singh VP, Kharya G, Mabalirajan U. *Obesity impairs cardiolipin-dependent mitophagy and therapeutic intercellular mitochondrial transfer ability of mesenchymal stem cells*. *Cell Death & Disease*. 2023 May 13;14(5):324.
- Faizan MI*, Chaudhuri R, Sagar S, Albogami S, **Chaudhary N**, Azmi I, Akhtar A, Ali SM, Kumar R, Iqbal J, Joshi MC. *NSP4 and ORF9B of SARS-CoV-2 induce pro-inflammatory mitochondrial DNA release in inner membrane-derived vesicles*. *Cells*. 2022 Sep 23;11(19):2969.
- Ahmad T*, Vullhorst D, Chaudhuri R, Guardia CM, **Chaudhary N**, Karavanova I, Bonifacino JS, Buonanno A. *Transcytosis and trans-synaptic retention by postsynaptic ErbB4 underlie axonal*

accumulation of NRG3. Journal of Cell Biology. 2022 May 17;221(7):e202110167.

- Azmi I*, Faizan MI, Kumar R, Raj Yadav S, **Chaudhary N**, Kumar Singh D, Butola R, Ganotra A, Datt Joshi G, Deep Jhingan G, Iqbal J. *A saliva-based RNA extraction-free workflow integrated with Cas13a for SARS-CoV-2 detection. Frontiers in cellular and infection microbiology.* 2021 Mar 16;11:632646.
- Sagar S*, Kapoor H, **Chaudhary N**, Roy SS. *Cellular and mitochondrial calcium communication in obstructive lung disorders. Mitochondrion.* 2021 May 1;58:184-99.

Skills

Technical Skills

- **Deep Learning & AI:** CNNs, attention-based MIL, transformers for WSI analysis; classification, detection, and multimodal modeling
- **Frameworks:** PyTorch, TensorFlow, Keras
- **Computational Pathology:** WSI preprocessing, patch extraction, feature embedding (ResNet), attention heatmaps; OpenSlide, OpenCV, QuPath
- **Programming:** Python, R; shell scripting
- **Spatial & Omics:** Scanpy, Squidpy, AnnData (single-cell & Visium data analysis)
- **Data Analysis & Visualization:** NumPy, pandas, Matplotlib; UMAP, t-SNE, ROC/PR
- **HPC & Reproducibility:** HPC environments, workflow automation

Research Skills

- Experimental design, cohort construction, and hypothesis-driven analysis
- Large-scale WSI dataset curation and multi-institutional data integration
- Scientific writing: manuscripts, grants, patents; conference presentations

Soft Skills

- Led model development and data pipelines for computational pathology projects
- Collaborated with pathologists and interdisciplinary research teams
- Mentored junior researchers in ML workflows and experimentation

Patents

- Ahmad T, **Chaudhary N**. An artificial intelligence-based system for prognosis and stage-specific diagnosis of oral cancer. **Provisional Indian Patent 202211005720**. February 2022.
- **Chaudhary N**, Ahmad T, Chaurasia A, Rai A, Mishra D. System and method for genomic markers and digital pathology image-based prediction of oral malignant disorders. **GOI Indian Patent No. 556810**. December 2024.

Awards & Honors

- Travel Grant Award - European Congress on Digital Pathology 2026, Graz, Austria (June 2026)
- Travel Grant Award - Asian Society of Digital Pathology 2025, Mumbai, India (October 2025)
- Senior research fellowship by the Indian Council of Medical Research (ICMR). 2022-2025

Selected presentations _____

- Accepted abstract for oral presentation at “**European Congress of Digital Pathology**”, under ‘AI Agents’ session, 22nd annual congress of ESDIP, in Graz, Austria (2026)
- Podium Presentation at **Pixels to Practice: Advancing Digital Pathology in Asia**, 2nd annual congress of the Asian Society of Digital Pathology, Mumbai, India (2025)
- **Grade-level classification of oral squamous cell carcinoma (OSCC) from digital pathology using ensemble deep learning algorithms**. Oral presentation, 15th RECOMB satellite workshop in Computational Cancer Biology, Istanbul, Turkey (2023)
- **An artificial intelligence based oral cancer diagnosis system**. Poster presentation, The role of AI in transforming healthcare, Jointly organized by THSTI-Faridabad and Oxford University, UK (2022)
- **Histology image based diagnostic tool for Oral Submucous Fibrosis (OSMF)**. Oral presentation, BIOPHYSIKA, Jamia Millia Islamia, New Delhi, India (2019)

References

1. **Dr. Tanveer Ahmad**
Director & Co-founder, Cellogen Therapeutics Pvt. Ltd.
Noida, Uttar Pradesh, India.
Ex-Assistant Professor,
Multidisciplinary Centre for Advanced Research & Studies (MCARS),
Jamia Millia Islamia, New Delhi, India.
Email: tanvir.333@gmail.com
2. **Dr. Sowmya SV**
Professor & Head,
Dept. of Oral & Maxillofacial Pathology & Oral Microbiology,
Oral Cancer Research Centre, Faculty of Dental Sciences,
Ramaiah University of Applied Sciences,
Bangalore, India.
Email: drsowmya25@gmail.com
3. **Dr. Sivaprakash Ramalingam**
Associate Professor,
Department of Biological Sciences and Bioengineering,
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Indian Institute of Technology Kanpur, Uttar Pradesh, India
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4. **Dr. Anurag Agarwal**
Dean, Biosciences and Health Research,
Trivedi School of Biosciences, Ashoka University,
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